INTERNSHIP AND PLACEMENT TRACKER

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "INTERNSHIP AND PLACEMENT TRACKER" is the bonafide work of DHARINI BALAMURUGAN (241501044), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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EXTERNAL EXAMINER

DECLARATION

I, DHARINI BALAMURUGAN (241501044), hereby declare that the project report titled “DESIGN AND DEVELOPMENT OF AN INTERNSHIP AND PLACEMENT TRACKER USING ZOHO CREATOR”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date:

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ACKNOWLEDGEMENT

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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

DHARINI BALAMURUGAN

ABSTRACT

Internships and campus placements are important stages in a student's career, and a college has to keep track of students, their skills, recruiting companies, internship and placement openings, applications, interviews and completed internships. When this information is kept in spreadsheets, e-mails and separate documents, it becomes scattered and duplicated, and it is difficult to see the status of an application, to coordinate interviews or to know whether an internship has been completed. A centralised and controlled application is therefore needed to organise these activities and to improve coordination among students, placement coordinators, HR administrators, career counsellors and companies.

This project presents the Internship and Placement Tracker, a low-code application developed on the Zoho Creator platform. The application is built around eight forms: Student Profile, Student Skill, Company Profile, Internship Opportunity, Placement Opportunity, Application Record, Interview Schedule and Internship Record. Lookup-style reference fields connect applications, interviews and internship records to the students and opportunities concerned. Fifteen report configurations group the records by status, department, interview round and skill proficiency. A dashboard page shows stipend and salary totals, average CGPA values, application-status indicators, a chart of internship durations, quick links and an embedded application report. A Deluge form workflow fills and disables the e-mail field when a student profile is added, and user permission profiles define access for the different roles.

A set of seventeen test cases has been prepared to verify the forms, the workflow, the relationships between records, the reports, the dashboard calculations and role-based access; their results, user acceptance feedback and deployment details are to be recorded from the live application. Automatic matching, notifications, controlled status transitions, placement analytics and audit logs are not built and are proposed as future enhancements. The project shows how a college can obtain a structured, maintainable placement-tracking system with a low-code platform and very little scripting.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Placement Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	INTERNSHIP AND PLACEMENT TRACKER
Application Domain	Education / Internship and Placement Management
Student Name & Register No.	DHARINI BALAMURUGAN – 241501044
Zoho Creator Edition	Education – Trial version
Application Name (Link Name)	INTERNSHIP AND PLACEMENT TRACKER
Published / Live URL	https://creatorapp.zoho.in/dharinibalamurugan.2024.a128/student-placement-management-system
Demo Credentials (per role)	Administrator, Placement Coordinator, HR Administrator, Career Counsellor, Student, Company Representative, System Analyst, Developer, Read, Write, Customer — use authorised demo accounts if required
Number of Forms / Subforms	8 / 0
Number of Reports (by type)	15 report configurations (8 general + 7 categorised); types:
Number of Pages / Dashboards	1 – DashBoard
Number of Workflows	Form: 1 (Auto-fill Student Email) Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	1 form workflow script (DS-01); no custom functions identified
Integrations	None configured. URL fields store resume, certificate, application and interview links; calendar and e-mail integrations are planned enhancements.
Roles & Profiles Defined	Administrator, Placement Coordinator, HR Administrator, Career Counselor, Student, Company Representative, System Analyst, Developer, Read, Write, Customer
Environments Used	Single environment – application developed and published directly
Date of Production Deployment	

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CGPA	Cumulative Grade Point Average
CSV	Comma-Separated Values
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FR	Functional Requirement
HR	Human Resources
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Internships and campus placements are important stages in a student's academic and professional development. A college therefore has to keep track of its students and their qualifications, the companies that recruit from the campus, the internships and job openings those companies announce, the applications that students submit, the interviews that follow, and the internships that students finally complete. These activities involve students, placement coordinators, HR administrators, career counsellors and companies.

1.2 Problem Statement

Educational institutions need an organised way to manage placement activities and internship programmes. Disconnected record-keeping makes it difficult to keep accurate student information, to match students with opportunities, to monitor the stages of recruitment, to schedule interviews and to track internship completion. A centralised system is therefore needed to store, update, retrieve and report placement-related information and to improve coordination among students, placement coordinators, HR administrators, career counsellors and companies. The specific problems are:

- Student, company and opportunity data is scattered across spreadsheets, e-mails and documents, so it is duplicated and inconsistent.
- The status of each application and its interview rounds cannot be seen in one place.
- Skills, certifications and internship progress are not recorded in a structured way.
- Staff cannot easily obtain summaries by status, department, interview round or skill level.

1.3 Objectives

- To design a structured data model of eight related forms for students, skills, companies, internships, placements, applications, interviews and internship records.

- To track each application and its interview rounds, results and feedback from submission to selection or rejection.
- To record internship completion status, mentor, dates and certificate links for each student.
- To provide 15 report configurations that group records by status, department, interview round and skill proficiency.
- To provide a dashboard with compensation and CGPA indicators, application-status indicators, an internship-duration chart and quick links.
- To apply role-based permission profiles and to automate the student e-mail field with a Deluge workflow.

1.4 Scope of the Project

In scope. The application contains eight forms, 15 report configurations, one dashboard page named DashBoard, one form workflow (Auto-fill Student Email) and a set of user permission profiles. The dashboard shows summary indicators, application-status indicators, an internship-duration chart, quick links and an embedded application report.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because the problem consists of structured records that are naturally represented as forms with related fields. Low-code application platforms provide visual tools for forms, reports and workflows so that developers can concentrate on the business requirement rather than on infrastructure and repetitive code [2], [3], [4]. Zoho Creator adds built-in hosting, storage, user permissions, a mobile-friendly interface and the Deluge scripting language for logic that cannot be configured visually [1], [7].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

At present, placement-related information in a college is usually kept in spreadsheets, e-mails and separate documents. The placement office records student details and resume links in one sheet, company contacts in another, and the internships and job openings announced by companies in a third. Applications are collected by e-mail or forms, and interview dates are communicated through messages. After an internship, the completion status and certificate are collected separately.

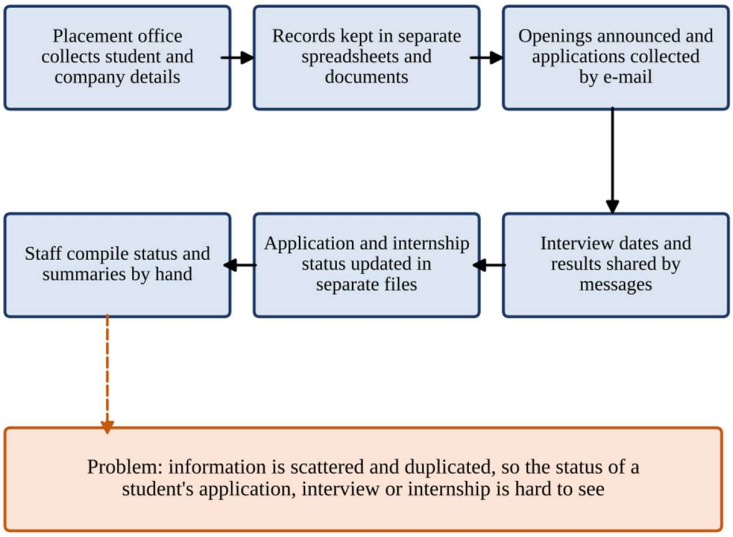


Figure 2.1: As-Is Process Flow of the Existing Internship and Placement Management Process

2.2 Limitations of the Existing System

- Information is spread across spreadsheets, e-mails and documents, so there is no single trusted record of a student's placement journey.
- Student, company and opportunity details are duplicated, which increases data-entry effort and inconsistency.
- Application status and interview outcomes are hard to monitor, because they are updated in different files.
- Skills, certifications and CGPA are not organised for quick review against eligibility criteria.

- Internship completion and certificate details are not tracked in a structured way.
- Producing a summary by status, department or interview round requires manual filtering and counting.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Needs separate design and coding of the database, back end, user interface and security; longest for an application with eight forms and many reports.	Forms, reports and pages are configured visually; the eight forms and their lookups can be built quickly, and Deluge is needed only for small pieces of logic.
Skills required	Front-end, back-end, database and deployment skills.	Understanding of data modelling and the platform; basic Deluge scripting.
Hosting & infrastructure	Server, database, domain and deployment pipeline must be arranged and maintained.	Hosting and storage are provided by the Zoho Creator platform, subject to the selected plan.
Mobile access	A separate responsive front end or native app would have to be developed.	The application can be opened in a mobile browser or the Zoho Creator mobile app without extra development.
Security & access control	Authentication, authorisation and session handling must be implemented.	Users and permission profiles are built in and configured, not coded.
Maintenance & change	A change to a field or report may touch several layers of code and need redeployment.	A new field, report or permission can usually be changed by the administrator in the builder.
Customisation limits	Almost unlimited design freedom.	Limited by the platform's field types, page builder and Deluge capabilities (see Section 8.4).
Cost	Developer time, hosting and maintenance costs.	Platform subscription according to the plan chosen; an education or trial plan may be used for learning.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation [1]	Zoho Creator Resource Center and Help (online)	Documents forms, field types, reports, pages, workflows, user management and publishing in Zoho Creator.	Reference for designing the eight forms, reports, dashboard and permission profiles.
2	R. Sanchis, Ó. García-Perales, F. Fraile and R. Poler, 2020 [2]	Low-Code as Enabler of Digital Transformation in Manufacturing Industry	Argues that low-code platforms facilitate and automate the development of enterprise applications and benchmarks existing platforms.	Academic support for using a low-code platform to digitise an institutional process.
3	A. C. Bock and U. Frank, 2021 [3]	Low-Code Platform	Characterises low-code platforms and discusses their features and implications for application development.	Defines the low-code platform concept used in Chapter 1.
4	D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi and M. Wimmer, 2022 [4]	Low-code development and model-driven engineering: Two sides of the same coin?	Examines the relationship between low-code development and model-driven engineering.	Explains why form and data models are the foundation of a low-code application.
5	R. Waszkowski, 2019 [5]	Low-code platform for automating business processes in manufacturing	Reports the use of a low-code platform to automate business processes.	Supports workflow-based automation of an administrative process.
6	R. Rokis and M. Kirikova, 2022 [6]	Challenges of Low-Code/No-Code Software Development: A Literature Review	Reviews the literature on the challenges of low-code and no-code development.	Informs the limitations discussed in Section 8.4.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
7	Zoho Corporation [7]	Introduction to Deluge – Zoho Deluge Help (online)	Describes the Deluge scripting language and its use for automation in Zoho applications.	Reference for the form-event workflow script DS-01 and the planned scripts.
8	Zoho Corporation [8]	Zoho Creator Academy (online)	Provides tutorials and training material for building Zoho Creator applications.	Used as a learning resource while building and reviewing the application.

2.5 Emerging Trends in Low-Code Development

Several trends in low-code development are relevant to this project. The first is citizen development: platforms allow people who are not professional programmers to build and maintain business applications, which suits a placement office that must keep the application running after the project is complete [2], [3]. The second is workflow automation: repetitive tasks are moved from people to the application, and the Auto-fill Student Email workflow is a small example [5].

2.6 Proposed System

The proposed Internship and Placement Tracker is a centralised application developed with Zoho Creator. It stores student profiles, skills, companies, internship and placement opportunities, applications, interviews and internship records in eight related forms; it provides 15 report configurations and a dashboard; and it applies role-based permission profiles for different users.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Student	Primary user whose profile, skills and applications are recorded.	Maintains personal information, skills, resume details and applications.
Placement Coordinator	Staff member who manages opportunities.	Creates and manages internship and placement opportunities and reviews related records.
HR Administrator	Staff member who handles recruitment records.	Manages application records and interview schedules.
Career Counsellor	Staff member who guides students.	Reviews student profiles, student skills and internship records (view access).
Company Representative	Representative of a recruiting organisation.	Maintains company information and takes part in recruitment-related activities.
Administrator	Application administrator.	Administers the application with broad access; other profiles (System Analyst, Developer, Read, Write, Customer) are also defined in the export.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain student profiles with student ID, register number, name, e-mail, mobile number, department, year, CGPA, graduation year and resume URL.	Student, Administrator	High
FR-02	The system shall automatically fill and disable the e-mail field with the logged-in user's e-mail when a student profile is added.	Student	Medium
FR-03	The system shall record student skills with proficiency level and certification details.	Student, Career Counsellor	High
FR-04	The system shall maintain company profiles with contact, address and location details.	Company Representative	High

FR ID	Requirement	Role(s)	Priority
FR-05	The system shall maintain internship opportunities with work mode, stipend, duration, eligibility, deadline and status.	Placement Coordinator	High
FR-06	The system shall maintain placement opportunities with required skills, minimum CGPA, salary package, job type, deadline and status.	Placement Coordinator	High
FR-07	The system shall record applications that link a student to an internship or placement opportunity with an application status.	Student, HR Administrator	High
FR-08	The system shall schedule interviews against an application with round, date, time, mode, result and feedback.	HR Administrator	High
FR-09	The system shall maintain internship records with dates, mentor, completion status, certificate URL and feedback.	Career Counsellor, Administrator	Medium
FR-10	The system shall provide report configurations for all main forms and grouped by status, department, interview round and skill proficiency.	All roles	High
FR-11	The system shall display dashboard indicators for stipend and salary totals, average CGPA values and application-status counts.	Placement Coordinator	Medium
FR-12	The system shall display an internship-duration chart, quick links to add records and an embedded application report on the dashboard.	Placement Coordinator, HR Administrator	Medium
FR-13	The system shall define user permission profiles that control access to modules.	All roles	High
FR-14	The system shall link related records through reference (lookup) fields for student, company, internship, placement and application.	Administrator	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Forms and reports shall be clear enough for placement staff and students to enter and find records with minimal training.
NFR-02	Performance	Forms, reports and the dashboard shall load within an acceptable time under normal use.
NFR-03	Security	Users shall access only the modules and actions permitted by their permission profile; record-level restrictions shall be tested before real student data is used.
NFR-04	Maintainability	Forms, reports and permissions shall be configurable by the administrator without reprogramming.
NFR-05	Accessibility / Usability on devices	The application shall be usable in supported browsers and on mobile devices through a browser or the Zoho Creator mobile app.
NFR-06	Data integrity	Related records shall refer to existing students, companies and opportunities through lookup fields; predefined choices shall limit status values.
NFR-07	Reliability	Records shall remain consistent across related forms and reports.
NFR-08	Extensibility	Additional forms, workflows and reports shall be addable as institutional requirements grow.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not used (no external API is called)
Other Zoho services	None configured; Zoho e-mail notification and Google integrations are future enhancements
Mobile	Zoho Creator mobile app (Android / iOS) or a mobile browser
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

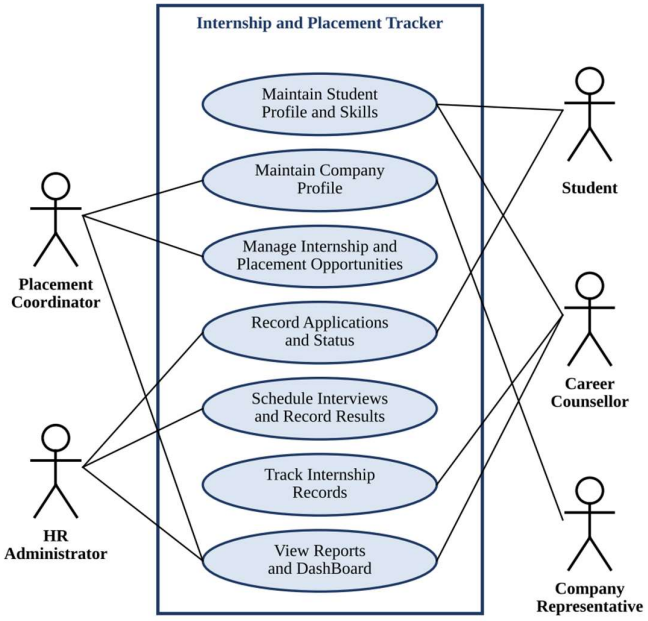


Figure 3.1: Use Case Diagram of the Internship and Placement Tracker

Table 3.5: Use Case Description – Register Student Profile and Skills

Use Case ID	UC-01
Use Case Name	Register Student Profile and Skills
Actor(s)	Student
Pre-condition	The student can sign in to the application and has a role that allows adding a profile.
Main Flow	1. The student opens the Student Profile form. 2. The application fills the e-mail field with the logged-in user's e-mail and disables it (workflow DS-01). 3. The student enters the register number, name, mobile number, department, year, CGPA, graduation year and resume URL, and submits the form. 4. The student opens the Student Skill form, selects the profile, and enters the skill, proficiency level and certification details.
Alternate Flow	A required value is missing or invalid; the application does not accept the record and the student corrects the entry.
Post-condition	The student profile and skills are stored and can be referred to by applications and internship records.
Related FR	FR-01, FR-02, FR-03

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

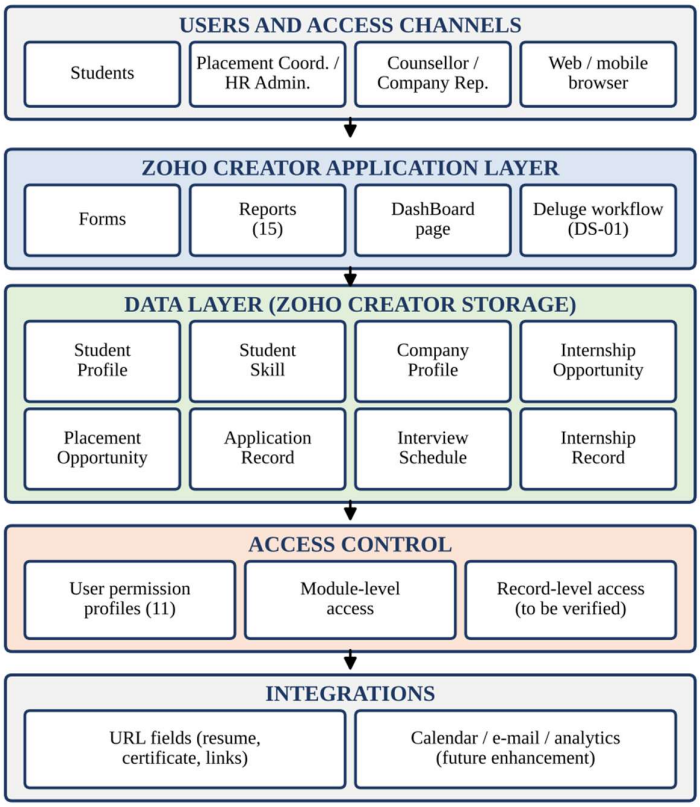


Figure 4.1: System Architecture of the Internship and Placement Tracker

The application follows a layered architecture on Zoho Creator, shown in Figure 4.1. The top layer contains the users – students, placement coordinators, HR administrators, career counsellors, company representatives and the administrator – who reach the application through a web or mobile browser. The application layer contains the forms that capture data, the reports that list and group records, the DashBoard page and the Deluge workflow.

4.2 Data Model

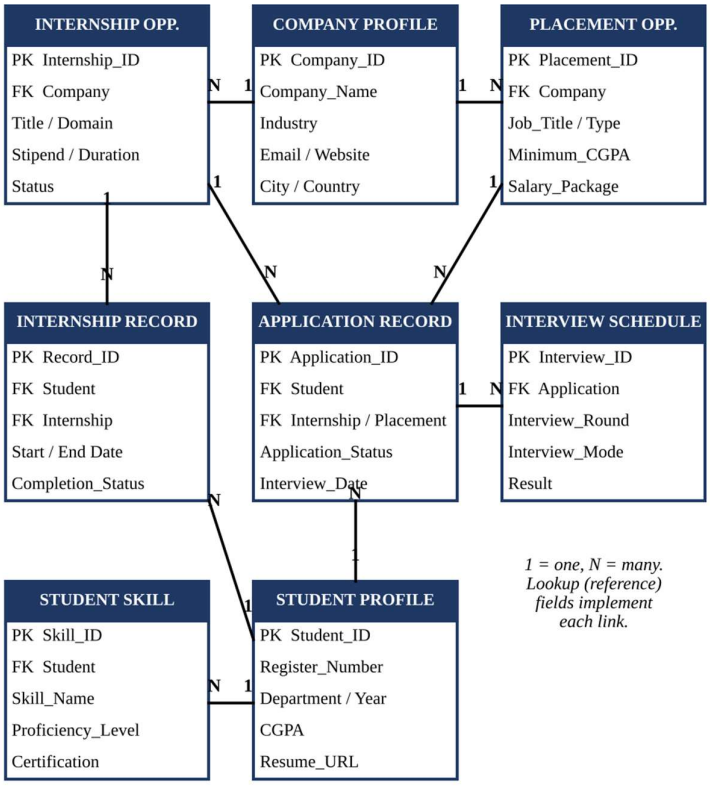


Figure 4.2: Entity–Relationship Diagram of the Internship and Placement Tracker

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Student Profile	Student_ID, Register_Number, Full_Name, Email, Department, CGPA, Resume_URL	Student Skill, Application Record, Internship Record	One-to-Many	Lookup field Student in the three forms
Student Skill	Skill_ID, Student, Skill_Name, Proficiency_Level, Certificate_URL	Student Profile	Many-to-One	Lookup field Student
Company Profile	Company_ID, Company_Name, Industry, Email, Website, Location	Internship and Placement Opportunity	One-to-Many	Lookup field Company in both forms

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Internship Opportunity	Internship_ID, Title, Company, Work_Mode, Stipend, Duration, Status	Company; Application; Internship Record	Many-to-One; One-to-Many	Lookup Company; referred to by applications and records
Placement Opportunity	Placement_ID, Job_Title, Company, Minimum_CGPA, Salary_Package, Status	Company; Application Record	Many-to-One; One-to-Many	Lookup Company; referred to by applications
Application Record	Application_ID, Student, Opportunity_Type, Application_Status	Student; opportunities; Interview Schedule	Many-to-One; One-to-Many	Lookups Student and Internship / Placement
Interview Schedule	Interview_ID, Application, Interview_Round, Result	Application Record	Many-to-One	Lookup field Application
Internship Record	Record_ID, Student, Internship, Start_Date, Completion_Status	Student; Internship Opportunity	Many-to-One	Lookups Student and Internship

Student Profile and Company Profile are the master forms. Opportunity, application, interview and internship forms are transactional forms that refer to them, which avoids retyping student and company details. Link names follow the Zoho convention of replacing spaces with underscores and must be checked in the form builder. No subform is identified in the export.

4.3 Form Design

Table 4.2: Field Specification – Student Profile Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student ID	Student_ID	Auto Number / ID	Yes	Automatically generated; unique	Identifies the student
Register Number	Register_Number	Single Line	Yes	Should be unique; validate format	College register number

Full Name	Full_Name	Single Line	Yes	Cannot be blank	Student name
Email	Email	Email	Yes	Auto-filled with logged-in user's email; disabled	Workflow DS-01
Mobile Number	Mobile_Number	Phone	Yes	Validate phone number format	Contact number
Department	Department	Drop Down	Yes	Select from predefined department options	Department report
Year of Study	Year_of_Study	Drop Down	Yes	Select valid year of study	Academic year
CGPA	CGPA	Decimal	Yes	Validate range, e.g. 0–10	Dashboard average
Graduation Year	Graduation_Year	Number	Yes	Validate a reasonable graduation year	Year of graduation
Resume URL	Resume_URL	URL	No	Must be a valid URL if entered	Link to the resume

Table 4.3: Field Specification – Student Skill Form

Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Skill ID	Skill_ID	Auto Number / ID	Yes	Automatically generated; unique	Identifies the skill record

Student	Student	Looku p	Yes	Select a valid student record	Links the skill to a student
Skill Name	Skill_Name	Single Line	Yes	Cannot be blank	Technical skill
Proficiency Level	Proficiency_Level	Drop Down	Yes	Beginner, Intermediate , Advanced, Expert	Proficienc y report
Certificatio n Name	Certification_Nam e	Single Line	No	Optional; enter certification name if available	Certificati on obtained
Issuing Organisati on	Issuing_Organisati on	Single Line	No	Optional; enter issuing organisation	Issuer of the certificate
Completio n Date	Completion_Date	Date	No	Must be a valid date	Date of completion
Certificate URL	Certificate_URL	URL	No	Must be a valid URL if entered	Certificate link

Table 4.4: Field Specification – Company Profile Form

Field Label	**Link Name**	**Field Type**	**Mandator y**	**Validation / Default**	**Remarks**
Company ID	Company_ID	Auto Numbe r	Yes	Automaticall y generated; unique	Identifies the company
Company Name	Company_Nam e	Single Line	Yes	Cannot be blank	Recruiting organisatio n
Industry	Industry	Single Line	Yes	Cannot be blank	Business sector

Description	Description	Multi Line	No	Optional text	About the company
Email / Contact Number	Email, Contact_Number	Email / Phone	Yes	Validate email and phone format	Contact details
Website	Website	URL	No	Must be a valid URL if entered	Company website
Location / Address	Location, Address	Single Line / Multi Line	Yes	Location required; address may be optional	Location and complete address
City / State / Postal Code / Country	City, State, Postal_Code, Country	Single Line	Yes	Validate postal code format where applicable	Address components
Latitude / Longitude	Latitude, Longitude	Decimal	No	Latitude: −90 to 90; longitude: −180 to 180	Coordinates

Table 4.5: Field Specification – Internship Opportunity Form

Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Internship ID	Internship_ID	Auto Number	Yes	Automatically generated; unique	Identifies the internship
Internship Title	Internship_Title	Single Line	Yes	Cannot be blank	Title of the opening
Company	Company	Lookup	Yes	Lookup to Company Profile	Offering company
Internship Domain	Internship_Domain	Single Line	Yes	Cannot be blank	Area of work

Description / Location	Description, Location	Multi Line / Single Line	Yes	Description and location should be provided	Details and place
Work Mode	Work_Mode	Drop Down	Yes	On-site, Hybrid, Remote	Mode of work
Stipend	Stipend	Number / Currency	No	Must be zero or greater	Dashboard total
Duration	Duration	Single Line	Yes	Enter internship duration, e.g. 3 months	Duration chart
Eligibility Criteria	Eligibility_Criteria	Multi Line	Yes	Specify eligibility requirements	Eligibility
Application Deadline	Application_Deadline	Date	Yes	Must be a valid date	Last date to apply
Internship Status	Internship_Status	Drop Down	Yes	Open, On-going, Closed, Completed	Opportunity status
Application Link	Application_Link	URL	No	Must be a valid URL if entered	External link

Table 4.6: Field Specification – Placement Opportunity Form

Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Placement ID	Placement_ID	Auto Number	Yes	Automatically	Identifies the opening

				generated; unique	
Job Title	Job_Title	Single Line	Yes	Cannot be blank	Title of the job
Company	Company	Lookup	Yes	Lookup to Company Profile	Recruiting company
Job Description	Job_Description	Multi Line	Yes	Cannot be blank	Details
Required Skills	Required_Skills	Multi Line	Yes	Specify required skills	Skills required
Minimum CGPA	Minimum_CGPA	Decimal	Yes	Validate range from 0 to 10	Dashboard average
Salary Package	Salary_Package	Number / Currency	Yes	Must be zero or greater; specify salary units	Dashboard total
Job Location	Job_Location	Single Line	Yes	Cannot be blank	Place of work
Job Type	Job_Type	Drop Down	Yes	Full-time, Part-time, Contract	Type of job
Application Deadline	Application_Deadline	Date	Yes	Must be a valid date	Last date to apply
Placement Status	Placement_Status	Drop Down	Yes	Open, On-going, Closed, Completed	Opportunity status
Application Link	Application_Link	URL	No	Must be a valid URL if entered	External link

Table 4.7: Field Specification – Application Record Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Application ID	Application_ID	Auto Number	Yes	Automatically generated; unique	Identifies the application
Student	Student	Lookup	Yes	Lookup to Student Profile	Applicant
Opportunity Type	Opportunity_Type	Drop Down	Yes	Placement, Internship	Kind of opportunity
Internship / Placement	Internship, Placement	Lookup	Yes, as applicable	Select the corresponding opportunity based on Opportunity Type	Corresponding opportunity
Application Date	Application_Date	Date	Yes	Default to current date where supported	Date submitted
Application Status	Application_Status	Drop Down	Yes	Submitted, Shortlisted, Rejected, Selected	Reports, dashboard
Interview Date	Interview_Date	Date	No	Must be a valid date if entered	Date of interview
Remarks	Remarks	Multi Line	No	Optional text	Notes

Table 4.8: Field Specification – Interview Schedule Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Interview ID	Interview_ID	Auto Number	Yes	Automatically generated; unique	Identifies the interview
Application	Application	Lookup	Yes	Lookup to Application Record	Application concerned
Interview Round	Interview_Round	Drop Down	Yes	Group Discussion, Technical, HR, Management	Round of interview
Interview Date / Time	Interview_Date, Interview_Time	Date / Time	Yes	Must be a valid date and time	Schedule
Interview Mode	Interview_Mode	Drop Down	Yes	Online, Offline, Hybrid	Mode
Interview Link	Interview_Link	URL	No	Required for online interviews; validate URL if entered	Link for online mode
Result	Result	Drop Down	Yes	Passed, Failed, On Hold	Outcome
Feedback	Feedback	Multi Line	No	Optional text	Interview feedback

Table 4.9: Field Specification – Internship Record Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Record ID	Record_ID	Auto Number	Yes	Automatically generated; unique	Identifies the record
Student	Student	Lookup	Yes	Lookup to Student Profile	Intern
Internship	Internship	Lookup	Yes	Lookup to Internship Opportunity	Internship undertaken
Start Date / End Date	Start_Date, End_Date	Date	Yes	End date must be on or after start date	Period of internship
Mentor Name	Mentor_Name	Single Line	No	Optional text	Mentor
Completion Status	Completion_Status	Drop Down	Yes	In Progress, Completed, Dropped	Status of the internship
Certificate URL	Certificate_URL	URL	No	Must be a valid URL if entered	Certificate link
Feedback	Feedback	Multi Line	No	Optional text	Feedback

4.4 Process and Workflow Design

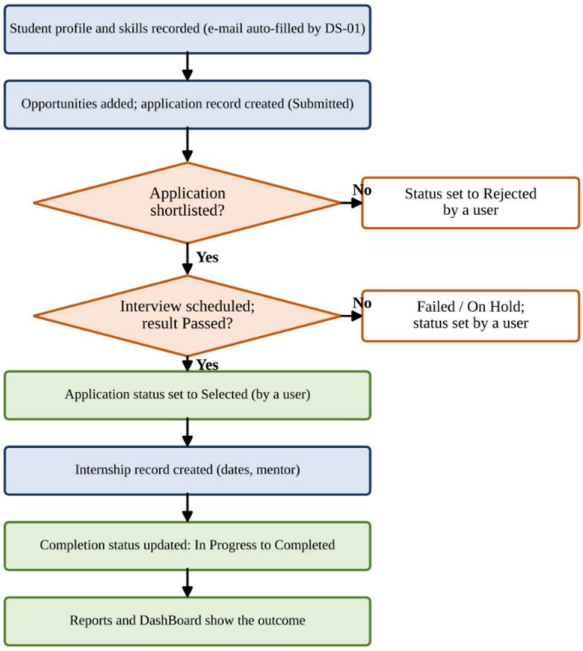


Figure 4.3: Process Flow for Application and Interview Tracking

Figure 4.3 shows the working principle of the application. A student profile and skills are recorded first; companies and their internship and placement opportunities are added; an application record links the student to an opportunity; interviews are scheduled and their results recorded; and the application status is then updated to Selected or Rejected. Selected internships are tracked through an internship record until completion. These status changes are made by users; the application does not update them automatically from interview results.

4.5 User Interface and Navigation Design

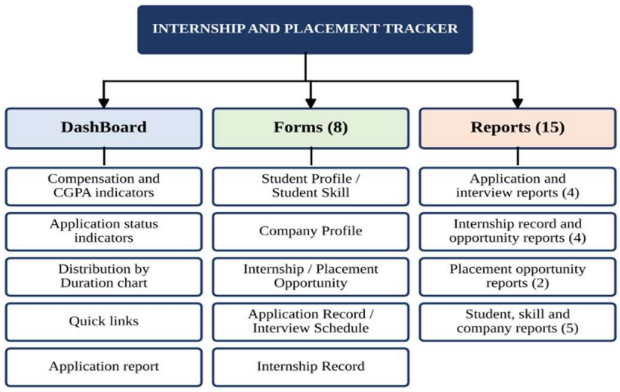


Figure 4.4: Logical Navigation Structure of the Internship and Placement Tracker

The application is organised around three groups of components: the DashBoard page, the eight forms, and the 15 report configurations. The DashBoard shows summary indicators, application-status indicators, the Distribution by Duration chart, quick links to add application records, internship records, interview schedules and internship opportunities, and an embedded application report.

4.6 Access Control Design

Table 4.10: Role–Permission Matrix

Component	**Administrator**	**Placement Coordinator**	**HR Administrator**	**Career Counsellor**	**Student**
Form: Student Profile	Broad	View / Manage	View	View	Maintain own data
Form: Student Skill	Broad	View / Manage	View	View / Manage	Maintain own data
Form: Company Profile	Broad	View	View / Manage	View	View
Form: Internship Opportunity	Broad	Create / Manage	View	View	View
Form: Placement Opportunity	Broad	Create / Manage	View	View	View
Form: Application Record	Broad	Review	Access	View	Maintain own data
Form: Interview Schedule	Broad	View	Access / Manage	View	View own interviews

Form: Internship Record	Broad	Review	View	View / Manage	View own records
Reports and Dashboard	Broad	View	View	View	View own relevant data

V = View. Broad = general administrative access. Access = access to the module, as described for the profile; the exact add / edit / view / delete rights, and the rights of the Company Representative, System Analyst, Developer, Read, Write and Customer profiles, are permission profiles of the application. The summary states the responsibilities of each profile but not every permission, so unstated cells are left as placeholders instead of being assumed.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in Zoho Creator under the name Internship and Placement Tracker. The first step was to identify the entities of the problem – students, skills, companies, internship opportunities, placement opportunities, applications, interviews and internship records – and to create one form for each, as designed in Chapter 4. Zoho Creator provides default reports for forms, and the 15 report configurations and the DashBoard page were built on these forms. Platform documentation and tutorials were used as references [1], [8].

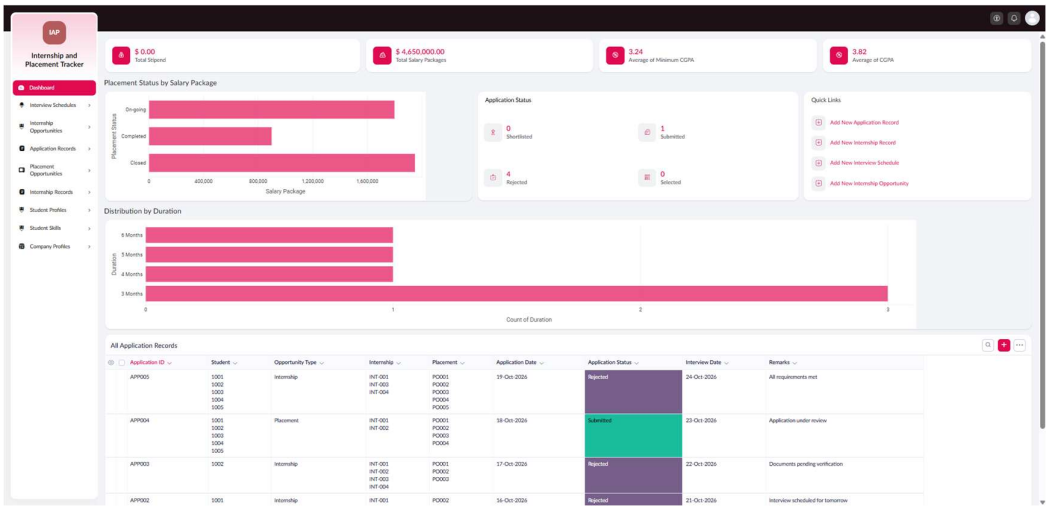


Figure 5.1: Internship and Placement Tracker – Application Home and Navigation Sections

5.2 Forms and Data Capture

Eight forms were configured with the fields listed in Tables 4.2 to 4.9. Choice fields carry predefined options – for example proficiency level, work mode, status, interview round, interview mode and result – and reference fields are lookup-style list fields that link a record to the student, company, opportunity or application it concerns. This reduces repeated entry of the same details.

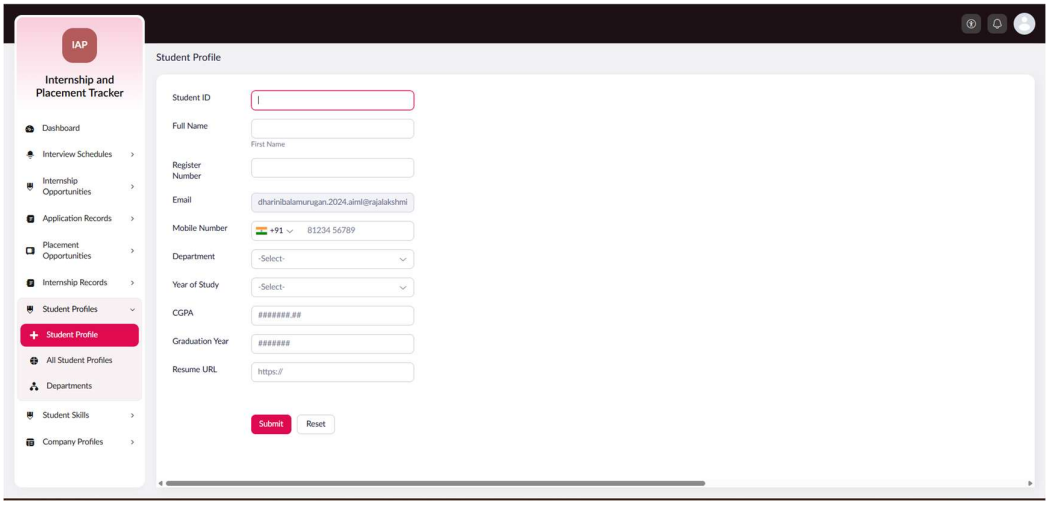


Figure 5.2: Student Profile Form in the Form Builder

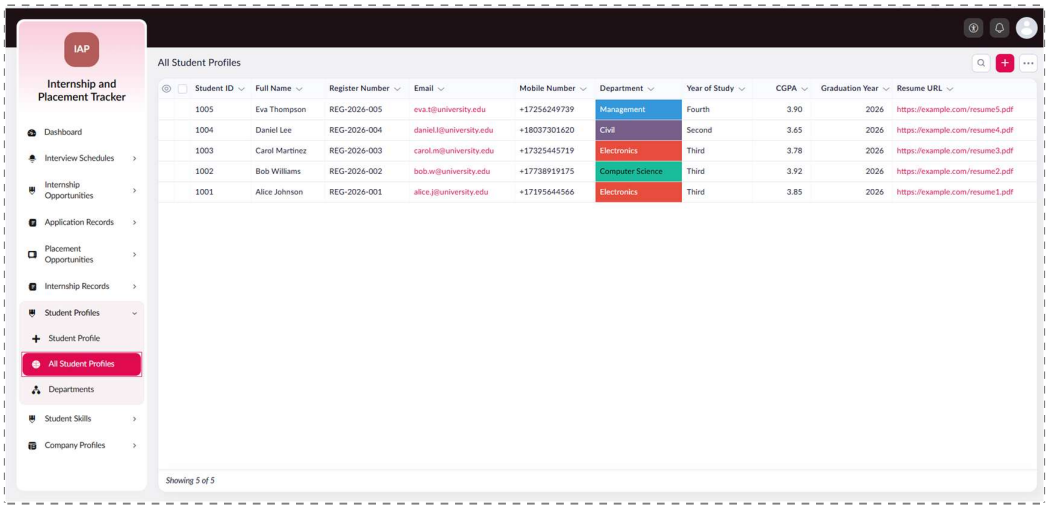


Figure 5.3: Student Profile Form in Live Mode

Student Skill form. This form records each skill with its proficiency level (Beginner, Intermediate, Advanced or Expert), certification name, issuing organisation, completion date and certificate URL, and refers to the student through a lookup field.

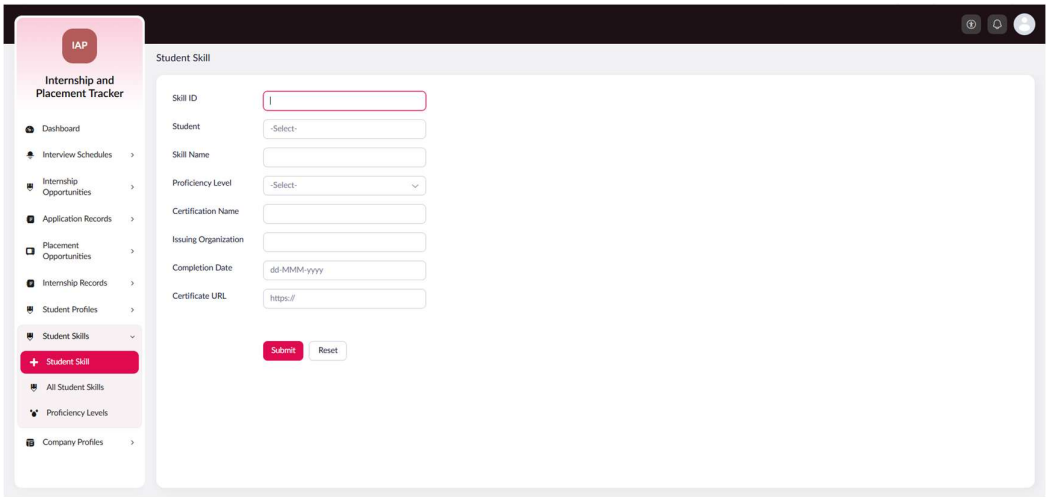


Figure 5.4: Student Skill Form in the Form Builder

Company Profile form. This form stores the company name, industry, description, contact details, website, location, full address and coordinates of each recruiting organisation.

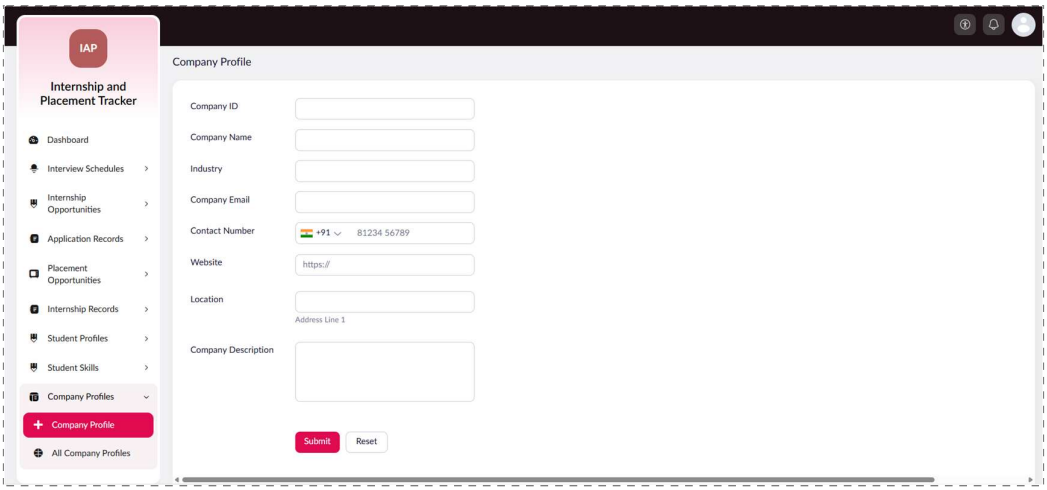


Figure 5.5: Company Profile Form in the Form Builder

Internship Opportunity form. This form records the title, company, domain, description, location, work mode (On-site, Hybrid or Remote), stipend, duration, eligibility, deadline, status (Open, On-going, Closed or Completed) and application link.

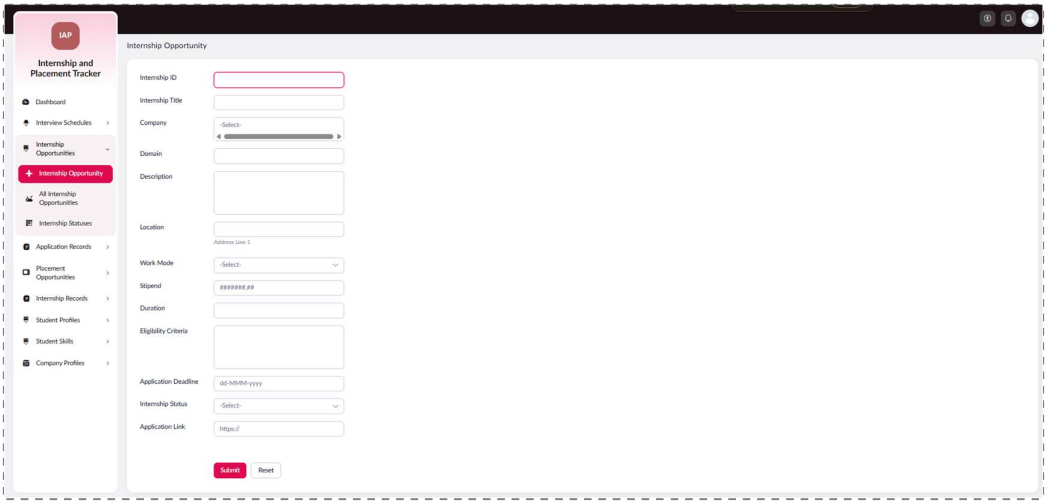


Figure 5.6: Internship Opportunity Form in the Form Builder

Placement Opportunity form. This form records the job title, company, description, required skills, minimum CGPA, salary package, location, job type (Full-time, Part-time or Contract), deadline, status and application link.

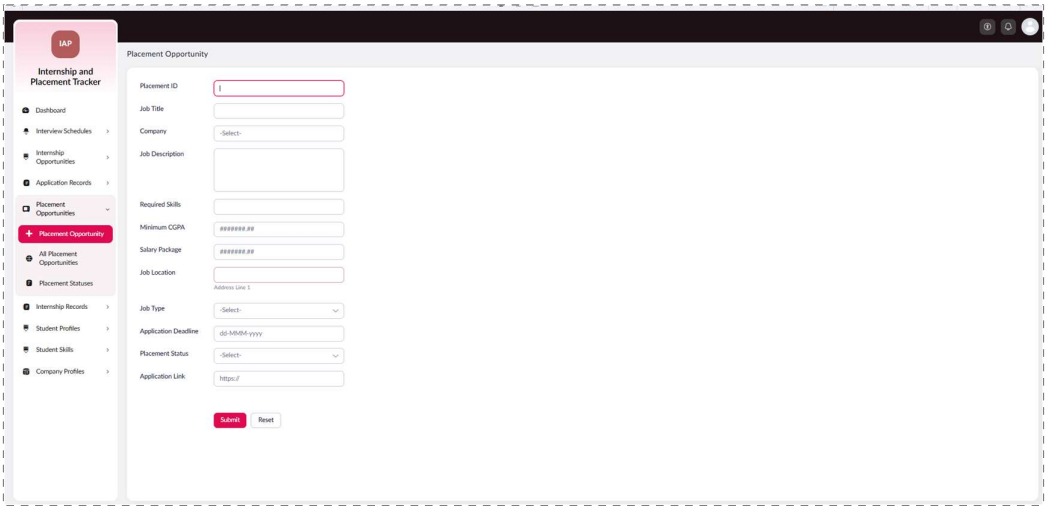


Figure 5.7: Placement Opportunity Form in the Form Builder

Application Record form. This form links a student to an internship or placement through the opportunity type and the corresponding reference, and records the application date, the status (Submitted, Shortlisted, Rejected or Selected), the interview date and remarks.

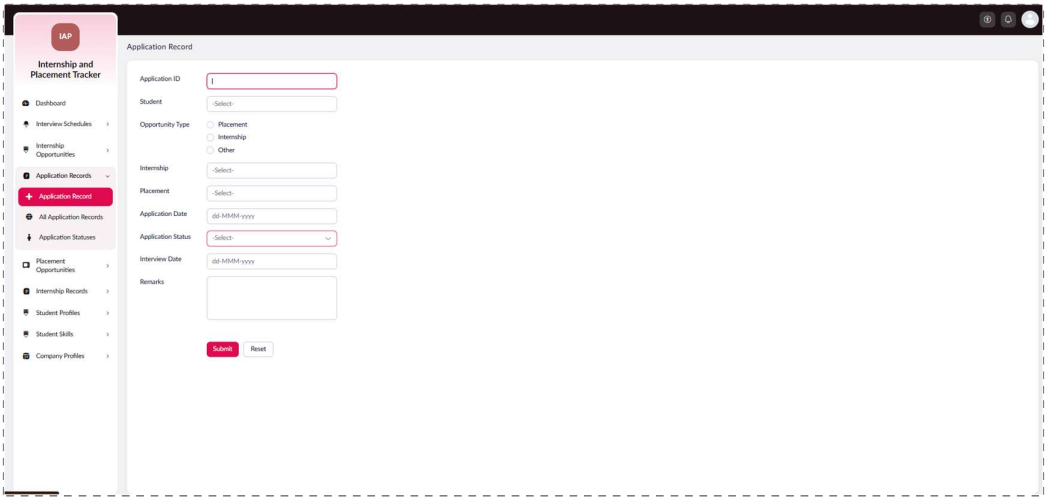


Figure 5.8: Application Record Form in the Form Builder

Interview Schedule form. This form refers to an application and records the round (Group Discussion, Technical, HR or Management), date, time, mode (Online, Offline or Hybrid), link, result (Passed, Failed or On Hold) and feedback.

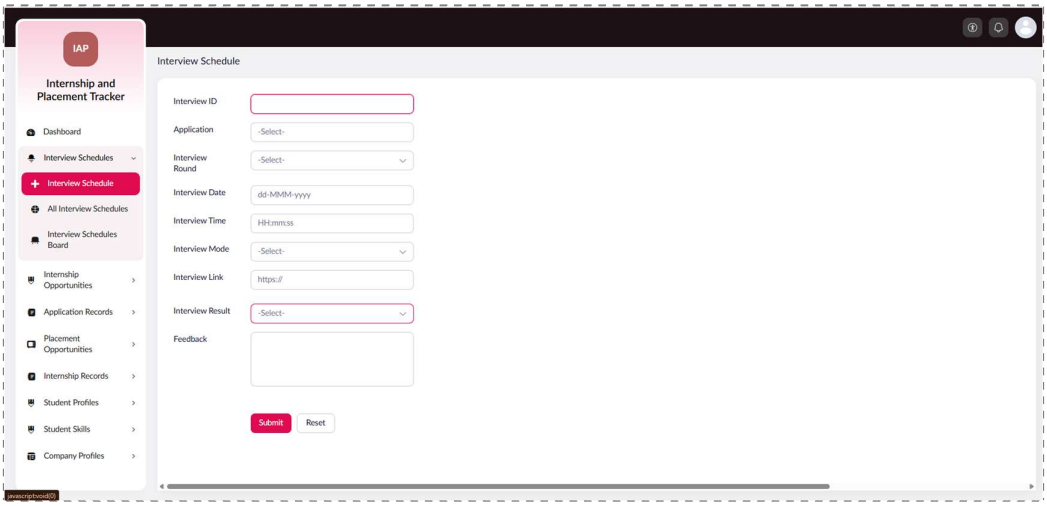


Figure 5.9: Interview Schedule Form in the Form Builder

Internship Record form. This form refers to the student and the internship and records the start and end dates, the mentor, the completion status (In Progress, Completed or Dropped), the certificate URL and the feedback.

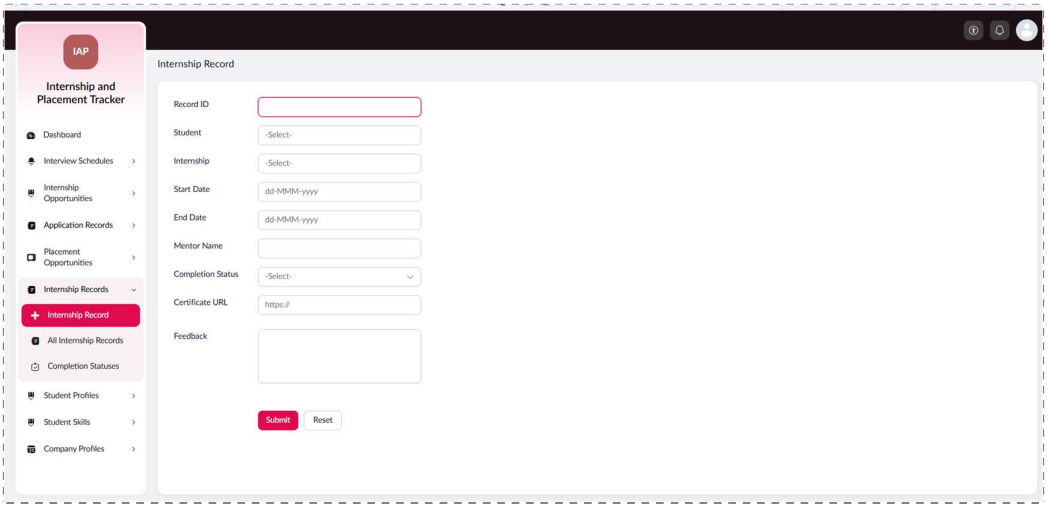


Figure 5.10: Internship Record Form in the Form Builder

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Application Record Report	List	Application Record	All records	View application details and status	HR, Placement
Application by Application Status	List	Application Record	By application status	Review applications by status	HR, Placement
Internship Record Report	List	Internship Record	All records	View internship records and progress	Counsellor, Admin
Internship by Completion Status	List	Internship Record	By completion status	Review completion status	Counsellor, Admin
Interview Schedule Report	List	Interview Schedule	All records	View interview schedules and results	HR

Interview by Interview Round	List	Interview Schedule	By interview round	Review interviews by round	HR
Internship Opportunity Report	List	Internship Opportunity	All records	View internship opportunities and details	Placement
Internship by Internship Status	List	Internship Opportunity	By internship status	Track internship opportunity status	Placement
Placement Opportunity Report	List	Placement Opportunity	All records	View placement opportunities and details	Placement
Placement by Placement Status	List	Placement Opportunity	By placement status	Track placement opportunity status	Placement
Student Profile Report	List	Student Profile	All records	View student profiles and academic details	Counsellor, Student
Student by Department	List	Student Profile	By department	Review students department-wise	Counsellor
Student Skill Report	List	Student Skill	All records	View student skills and proficiency	Counsellor, Student
Student Skill by Proficiency Level	List	Student Skill	By proficiency level	Review skills by proficiency level	Counsellor
Company Profile Report	List	Company Profile	All records	View company profiles and	Placement, Company Representative

				contact details	

Fifteen report configurations were defined: eight general reports (one for each form) and seven categorised reports. Report types, actions and the roles that can open each report are from the application; the Roles column shows the audience implied by the responsibilities of each profile and is not a permission setting. Reports turn individual records into information for review, for example the number of applications in each status or the distribution of students by department.

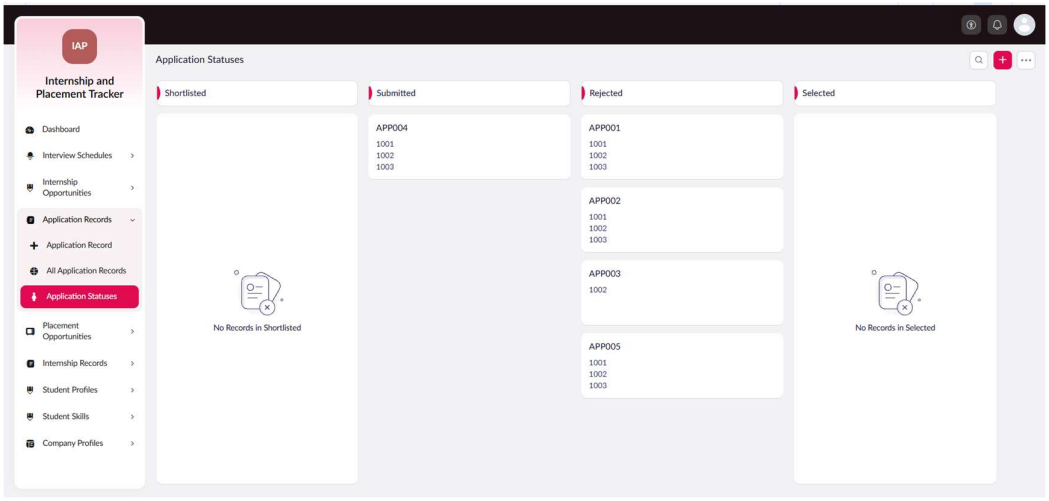


Figure 5.11: Application by Application Status Report

5.4 Pages and Dashboards

A page named DashBoard was built with the Zoho Creator page builder. It contains summary calculations for total internship stipend, total placement salary package, average minimum CGPA required for placements and average CGPA of registered students, together with four application-status indicators – Shortlisted, Submitted, Rejected and Selected – configured from the application-status field.

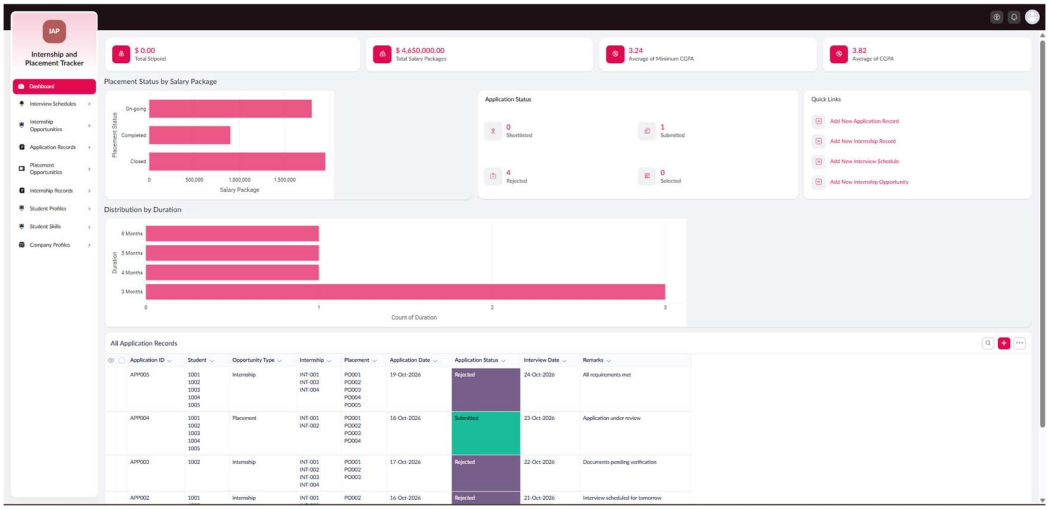


Figure 5.12: DashBoard Page

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Student Profile → On Load (Add) – workflow Auto-fill Student Email	Fill the e-mail field with the logged-in user's e-mail and disable it (implemented)	assignment, zoho.loginuserid, disable
DS-02	Internship Record → On Validate (planned)	Reject a record whose end date is before its start date (future enhancement, not implemented)	if, alert, cancel submit
DS-03	Interview Schedule → On Success (planned)	E-mail the candidate when an interview is scheduled (future enhancement, not implemented)	fetch, if, sendmail

Code 5.1: DS-01 – Auto-fill Student Email (implemented)

```
// DS-01 | Form: Student_Profile | Workflow: Auto-fill Student Email |
Event: On Load (Add)
// Purpose: Fill the e-mail field with the logged-in user's e-mail and
prevent manual editing
input.Email = zoho.loginuserid;
disable Email;
```

Code 5.2: DS-02 – Internship date validation (planned)

```
// DS-02 | PLANNED - not implemented in the exported application
// Form: Internship_Record | Event: On Validate
// Purpose: Reject an internship record whose end date is before its
start date
if(input.Start_Date != null && input.End_Date != null &&
input.End_Date < input.Start_Date)
{
    alert "End date cannot be before the start date.";
    cancel submit;
}
```

Code 5.3: DS-03 – Interview notification e-mail (planned)

```
// DS-03 | PLANNED - not implemented in the exported application
// Form: Interview_Schedule | Event: On Success
// Purpose: E-mail the candidate when an interview is scheduled
appRec = Application_Record[ID == input.Application];
stu = Student_Profile[ID == appRec.Student];
if(stu.Email != null)
{
    sendmail
    [
        from :zoho.adminuserid
        to :stu.Email
        subject :"Interview scheduled: " + input.Interview_Round
        message :"Date: " + input.Interview_Date + "<br/>Mode: " +
input.Interview_Mode
    ]
}
```

DS-01 is the only script confirmed in the exported application. It belongs to the workflow Auto-fill Student Email, which runs on the load event when a student profile is being added. It assigns zoho.loginuserid to the e-mail field and disables the field so that it cannot be edited during that form interaction. The export does not establish that the e-mail is separately verified as the student's institutional address.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Auto-fill Student Email	Form Workflow	On Load (record being added) of Student Profile	Always, on add	Assign zoho.loginuserid to the e-mail field and disable the field	DS-01
Other workflow types	Schedule / Approval / Blueprint / Batch	—	—	Not identified in the application; see Section 8.5	—

The export contains one explicitly configured form workflow. The inspected configuration contains no automation for opportunity matching, status transitions, interview notifications or certificate verification; application and interview status values are therefore set by users. Scheduled workflows, approvals and notifications are listed as future enhancements in Section 9.2.

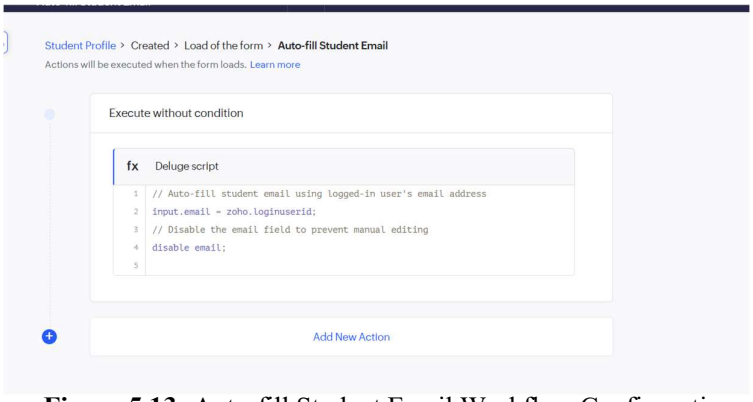


Figure 5.13: Auto-fill Student Email Workflow Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Link fields (URLs)	Resume, certificate, application and interview links stored as URL values	Stored in Student Profile, Student Skill, opportunity, interview and internship record forms	None (links are only stored)	No external call is made
External services (Google Calendar, Zoho Mail, Zoho Analytics, other)	Not configured	Future enhancement	—	Not part of the built application

No external integration is configured in the application. URLs can be stored in the resume, certificate, application and interview-link fields, but the application does not open, validate or verify them. No invokeurl script, webhook or API test is part of the project; calendar, notification and institutional-system integrations are described as future enhancements in Section 9.2.

5.8 User Management and Role-Based Access

The export defines eleven user profiles: Administrator, Placement Coordinator, HR Administrator, Career Counselor, Student, Company Representative, System Analyst, Developer, Read, Write and Customer. Their responsibilities are given in Table 3.1 and the permissions that are stated for them are summarised in Table 4.10. For example, the Career Counselor profile has view permissions for student profiles, student skills and internship records, the HR Administrator profile has access to application records and interview schedules, and the Placement Coordinator can create and manage internship and placement opportunities.

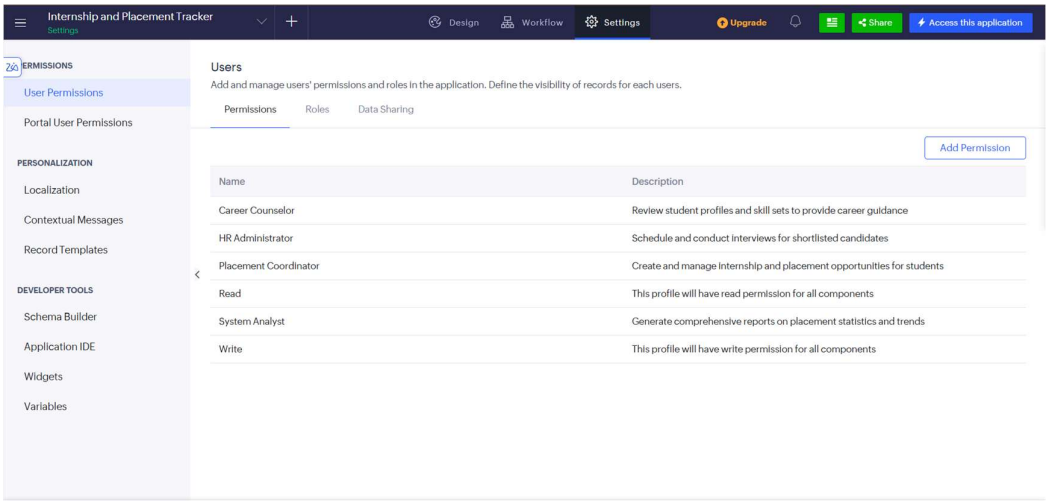


Figure 5.14: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

No Zia or other AI feature is part of the application, and no prediction or recommendation is implemented. Analytics are limited to the DashBoard: the stipend and salary totals, the two CGPA averages, the four application-status indicators and the Distribution by Duration bar chart. These give a quick overview of compensation, academic eligibility, application outcomes and internship durations.

5.10 Pro-Code Extension

A pro-code extension was not used. The forms, reports, dashboard, permission profiles and the single workflow script were achieved with configuration and one short Deluge script, so Zoho Catalyst, JavaScript widgets and custom APIs were not necessary.

5.11 Mobile and Portal Access

The application is a Zoho Creator web application and can be opened in a mobile browser or the Zoho Creator mobile app.



Figure 5.15: Internship and Placement Tracker on a Mobile Device

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is planned at four levels. Form testing checks that each of the eight forms saves valid records and that choice fields offer the configured options. Workflow testing checks the Auto-fill Student Email behaviour (DS-01). Integration testing checks that lookup references connect students, companies, opportunities, applications, interviews and internship records, and that the reports and dashboard calculations agree with the underlying records. Role-based testing checks that each user profile can reach only the permitted modules.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Add a student profile	Fill the Student Profile form and submit	Profile saved and displayed in Student Profile Report	Student profile saved successfully	Pass
TC-02	FR-02	E-mail auto-fill (DS-01)	Open Student Profile form in add mode	Email auto-filled with login email and disabled	Email auto-fill works correctly	Pass
TC-03	FR-03, FR-14	Add a student skill	Select a student, enter skill and proficiency level, and submit	Skill saved and linked to the student	Student skill saved and linked successfully	Pass
TC-04	FR-04	Add a company profile	Fill Company Profile form and submit	Company profile stored and selectable in	Company profile saved successfully	Pass

				opportunity forms		
TC-05	FR-05, FR-14	Add an internship opportunity	Select company, enter details, status, duration, and deadline	Internship details recorded correctly	Internship opportunity saved successfully	Pass
TC-06	FR-06, FR-14	Add a placement opportunity	Select company, enter job details, and submit	Placement details recorded and linked to company	Placement opportunity saved successfully	Pass
TC-07	FR-07, FR-14	Create an application	Select student, opportunity type, and opportunity, then submit	Application linked to the correct student and opportunity	Application created and linked successfully	Pass
TC-08	FR-07, FR-10	Update application status	Change status to Shortlisted and then Selected	Updated status appears in the status report	Application status updated successfully	Pass
TC-09	FR-08, FR-14	Schedule an interview	Select application and enter round, date, time, and mode	Interview schedule linked to the application	Interview schedule saved successfully	Pass
TC-10	FR-08	Record interview result	Set result and enter feedback	Result and feedback saved and shown in the report	Interview result and feedback saved successfully	Pass
TC-11	FR-09	Update internship completion	Set completion status to Completed	Record appears in the	Internship completion status	Pass

				completion status report	updated successfully	
TC-12	FR-10	Open grouped student reports	Open department and proficiency reports	Records appear under the correct groups	Grouped reports display records correctly	Pass
TC-13	FR-11	Check dashboard calculations	Compare totals and averages with underlying records	Dashboard values match the records	Dashboard calculations match the records	Pass
TC-14	FR-11	Check status indicators	Compare dashboard indicators with the status report	Counts match the records for each status	Status indicators display correct counts	Pass
TC-15	FR-12	Check chart and quick links	Open dashboard, inspect chart, and use quick links	Chart represents data and links open intended destinations	Chart and quick links work correctly	Pass
TC-16	FR-13, NFR-03	Test different user roles	Sign in with each profile and access modules	Only permitted modules and actions are accessible	Role-based access works as configured	Pass
TC-17	NFR-05	Open on a mobile device	Open the application in a mobile browser	Pages and controls are usable on a small screen	Application works correctly on a mobile screen	Pass

6.3 Debugging Deluge Scripts

The Deluge workflow DS-01 is a two-statement script. It can be checked by opening the Student Profile form in add mode and observing the e-mail field, and by using the info statement and the execution log of the platform if the value is not as expected.

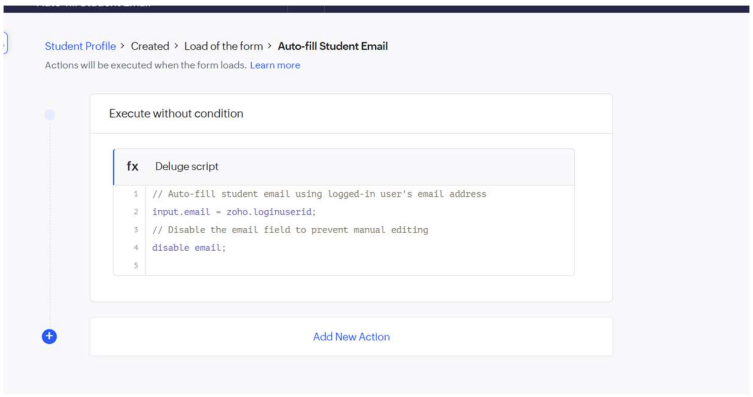


Figure 6.1: Deluge Debug Output / Log

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Student email field was not automatically populated when opening the Student Profile form.	Email auto-fill workflow was not correctly configured.	Configured the form-load workflow to populate the email using zoho.loginuserid and disable the field.	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	HR Administrator	Application records and interview schedules are organised and easy to review.	Reviewed the application and interview reports for clarity and usability.
User 2	Student	Student profiles, skill details, and application status information are useful and easy to access.	Reviewed the student forms and reports to improve ease of navigation.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

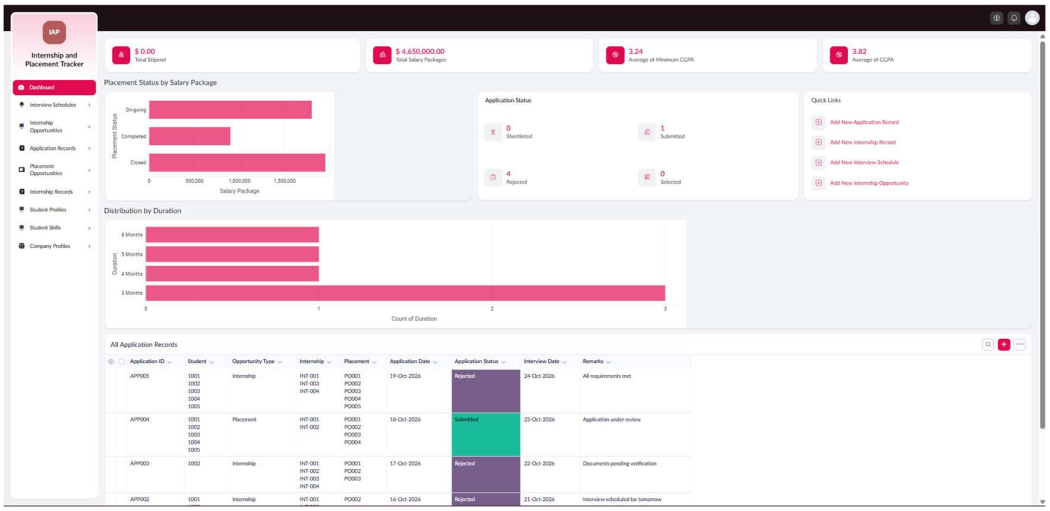


Figure 7.1: Environments / Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production	✓
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

7.2 Performance Optimisation

Candidate measures concern reports and the dashboard: show only needed columns, use grouped reports, and keep the dashboard light. No measurements were supplied, so Table 7.2 is left to be measured.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Show only required columns in large list reports	Student Profile Report, Application Record Report	Reports display more information than necessary, making records harder to review.	Display only essential fields such as student name, register number, department, application date, and application status.	Improves readability, reduces visual clutter, and makes important information easier to find.
Use grouped reports by status, department, and interview round for review	Grouped reports	Users need to review individual records to identify categories and compare information.	Use reports grouped by application status, internship completion status, student department, skill proficiency, and interview round.	Simplifies record analysis, speeds up reviews, and helps users identify trends and track progress.

7.3 Security Features

Security rests on Zoho Creator controls; the export does not by itself guarantee record-level restrictions, so these must be tested. The controls are listed in Table 7.3.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
User permission profiles (11 defined)	Forms, reports, and DashBoard	Unauthorised access to modules
Predefined choice fields	Status, mode, round, result, and proficiency fields	Inconsistent or invalid status values

Lookup (reference) fields	Student Skill, Application Record, Interview Schedule, and Internship Record forms	Incorrect references between related records
Email field disabled on add (DS-01)	Student Profile form	Accidental modification of the email value during entry
Record-level access and field visibility	Student records, applications, and related personal information	Unauthorised access to other students' personal data
Audit trail and change history	Application status updates and other record changes	Untraceable changes to application status

7.4 Governance and Administration

The administrator keeps data accurate and statuses current; personal data such as CGPA and resume links should be visible only to the roles that need it.

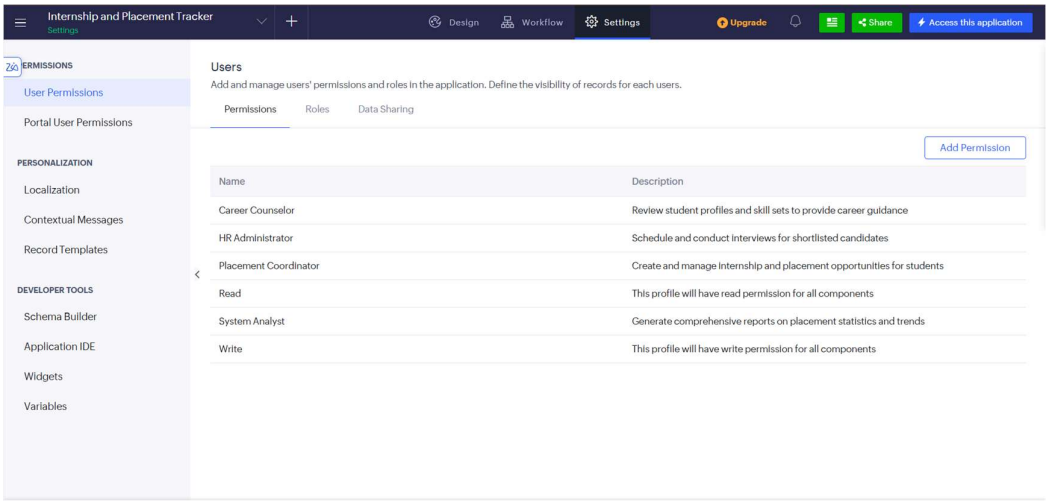


Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application is demonstrated with one realistic scenario that passes through the main roles. Step 1: a student adds a profile (the e-mail field is filled automatically) and records skills, while the placement coordinator adds a company with an internship and a placement opportunity. Step 2: an application is recorded for the student, an interview is scheduled in the Interview Schedule form, and the result and application status are updated. Step 3: the DashBoard and the status reports show the application outcomes, and an internship record tracks the internship until it is completed.

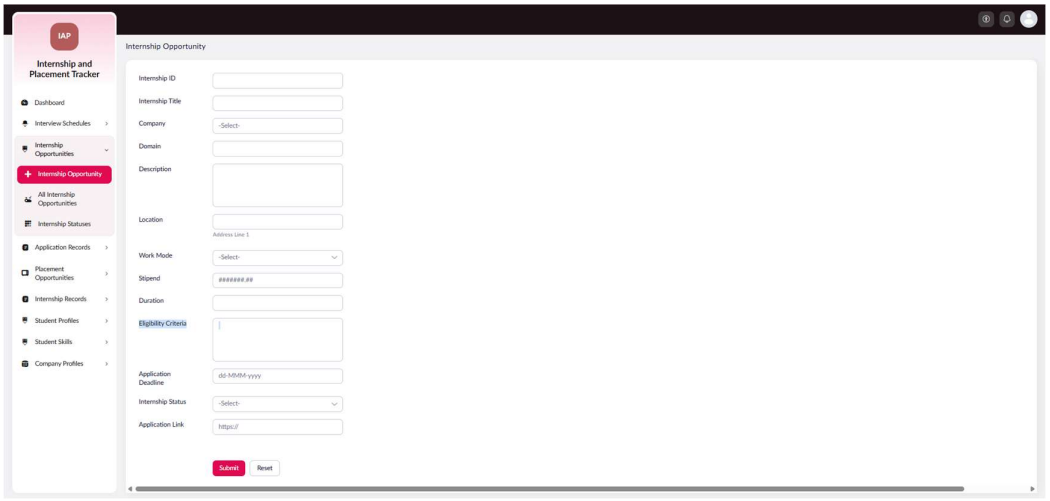


Figure 8.1: Scenario Step 1 – Student profile, skills and opportunities are recorded

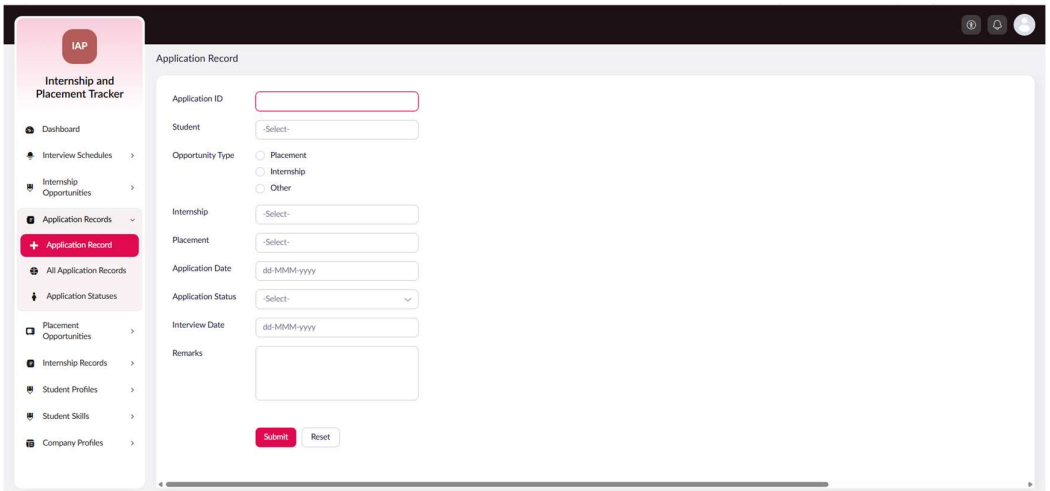


Figure 8.2: Scenario Step 2 – Application and interview are recorded

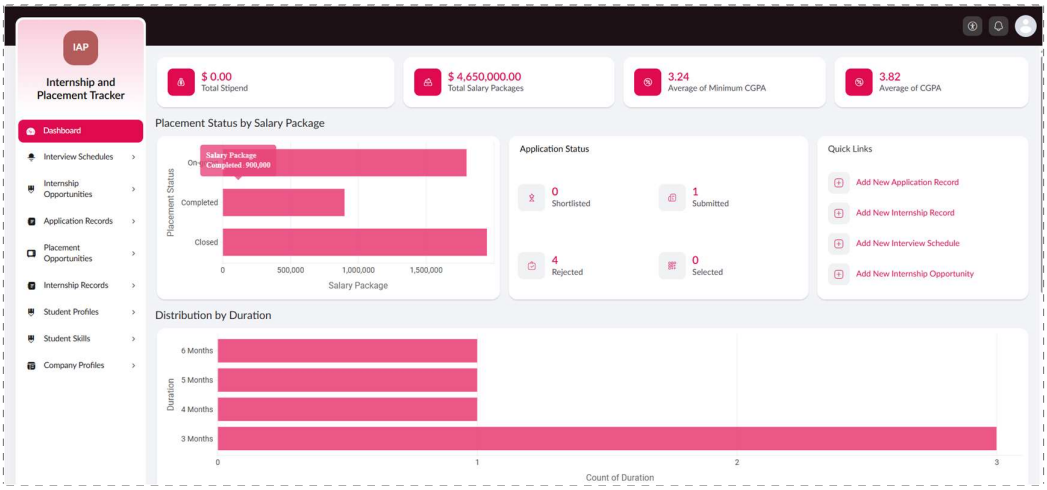


Figure 8.3: Scenario Step 3 – DashBoard and reports show the outcome

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To design a structured data model of eight related forms	Sections 4.2, 4.3, 5.2	Eight forms linked by lookup fields (Figures 4.2, 5.2–5.10; Tables 4.1–4.9)
To track applications and interviews	Sections 4.4, 5.2	Application Record and Interview Schedule forms (Figures 4.3, 5.8, 5.9)
To record internship completion	Sections 4.2, 5.2	Internship Record form and completion-status report (Figure 5.10)
To provide reports grouped by status, department, round and proficiency	Section 5.3	15 report configurations (Table 5.1, Figure 5.11)
To provide a dashboard with indicators, chart and quick links	Section 5.4	DashBoard page (Figures 5.12, 5.15)
To apply role-based permissions and automate the e-mail field	Sections 4.6, 5.5, 5.6, 5.8	Permission profiles, Table 4.10 and workflow DS-01 (Code 5.1, Figures 5.13, 5.14)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Record keeping	Spreadsheets, e-mails and documents	Eight related forms in one application	One structured source of placement information
Linking of records	Student and company details retyped	Lookup fields refer to existing records	Less repeated entry
Application and interview tracking	Status spread across files	Status fields and interview records linked to the application	Progress visible in one place
Report generation	Manual filtering and counting	15 report configurations grouped by status, department, round and proficiency	Repeatable reports
Summary view	Calculated by hand	DashBoard indicators, chart and quick links	Faster overview
Access control	Depends on who holds the files	User permission profiles	Controlled access, subject to testing

The comparison is qualitative. Quantitative values such as time saved, placement rates or user numbers were not measured and are not claimed.

8.4 Limitations of the Low-Code Approach

The constraints of the approach follow from the platform's boundaries. The page builder and report options are limited to the components provided by Zoho Creator, so a fully custom design would need a pro-code extension. Deluge and the platform have limits that matter for very large volumes. Some features, such as the number of users, environments, integrations and portal options, depend on the edition and plan [6].

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

On the spectrum from no-code to pro-code, nearly the whole application is no-code configuration: the eight forms, lookup fields, 15 reports, the DashBoard and the permission profiles are created in the builder without writing code. Low-code Deluge is used only for the one script that fills and disables the e-mail field.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Internship and Placement Tracker is a low-code application on Zoho Creator that centralises student career-related activities. Eight forms cover student profiles, skills, companies, internship opportunities, placement opportunities, applications, interviews and internship records; lookup fields connect them; 15 report configurations and a dashboard summarise the data; a workflow fills the student e-mail field; and several permission profiles define access for different users.

9.2 Future Enhancements

- Automated e-mail and in-app notifications for application deadlines and interview schedules.
- Skill-based internship and job recommendations that match student skills and CGPA with opportunity requirements.
- Placement statistics, including eligible students, students placed and department-wise outcomes.
- Resume upload and document validation in place of stored links.
- Calendar integration for interviews and internship milestones.
- Audit logs showing who changed an application status and when, and controlled status transitions.
- Improved validation for student IDs, duplicate applications, CGPA, dates and required fields.
- Additional filters for graduation year, company, department, salary package and opportunity status.

9.3 Learning Outcomes and Career Relevance

Developing this application gives practical experience of the low-code life cycle: analysing a placement-management problem, writing requirements, modelling data with forms and lookups, building reports and a dashboard, configuring permission profiles and a Deluge workflow, and planning tests. Designing the links between students, opportunities, applications and interviews shows the difference between master data and transactional data.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and sections/menu organised	Y	Internship and Placement Tracker application	5.1	Fig. 5.1
II	Forms with at least 8 distinct field types	Y	Eight main forms	5.2	Fig. 5.2–5.10; Tables 4.2–4.9
II	Lookup field (relationship between forms)	Y	Lookup references between related forms	4.2, 5.2	Fig. 4.2; Table 4.1
II	Subform (parent–child data)	N	No subform identified	4.2, 5.2	—
II	Field validations / layout rules	Y	Form fields and Student Email workflow	5.2	Table 4.2; Code 5.1
II	List report with filter, grouping, custom action	Y	Fifteen report configurations	5.3	Table 5.1; Fig. 5.11
II	Kanban / Calendar / Timeline / Map report	N	Not identified	5.3	—
II	Pivot table / chart report	N	No pivot report identified	5.3	—
II	Page / dashboard built with UI Builder	Y	DashBoard	5.4	Fig. 5.12
II	Deluge – form events (validate / success / field actions)	Y	Auto-fill Student Email on form load	5.5	Code 5.1

II	Deluge – functions, fetch, loops, collections	N	Not identified in the export	5.5	Code 5.2, 5.3 (planned)
II	Email / in-app notifications	N	Not identified	5.5, 5.6	—
II	Form workflow	Y	Auto-fill Student Email	5.6	Fig. 5.13; Table 5.3
II	Scheduled workflow	N	Not identified	5.6	—
II	Approval process	N	Not identified	5.6	—
II	Blueprint	N	Not identified	5.6	—
II	Batch / report workflow	N	Not identified	5.6	—
II	Testing and debugging evidence	N	Test cases documented; execution evidence not provided	6	Table 6.1; Fig. 6.1
III	Integration with a Zoho service	N	Not identified	5.7	—
III	Connection / REST API call (invokeurl)	N	Not identified	5.7	—
III	Webhook or Creator REST API	N	Not identified	5.7	—
III	Users, roles and permission profiles	Y	Eleven defined permission profiles	5.8	Fig. 5.14; Table 4.10
III	Record-level access criteria / customer portal	N	Not confirmed in the export	5.8	Section 5.8
III	Zia / AI feature	N	Not identified	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard chart and status indicators	5.9	Fig. 5.12, 5.15
IV	Performance optimisation applied and measured	N	Optimisation ideas documented; performance not measured	7.2	Table 7.2

IV	Deployment across environments	N	Deployment environments not confirmed	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	Permission profiles identified; other controls require verification	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Audit trail and governance settings not confirmed	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not identified	5.10	—
V	Low-code limitations and spectrum analysis	Y	Limitations and spectrum analysis	8.4, 8.5	Sections 8.4, 8.5
—	Mobile app / portal access	N	Not confirmed	5.11	Fig. 5.15

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 DS-01 Auto-fill Student Email (implemented)

```
// DS-01 | Form: Student_Profile | Workflow: Auto-fill Student Email |
Event: On Load (Add)
// Purpose: Fill the e-mail field with the logged-in user's e-mail and
prevent manual editing
input.Email = zoho.loginuserid;
disable Email;
```

C.2 DS-02 Internship date validation (planned)

```
// DS-02 | PLANNED - not implemented in the exported application
// Form: Internship_Record | Event: On Validate
// Purpose: Reject an internship record whose end date is before its
start date
if(input.Start_Date != null && input.End_Date != null &&
input.End_Date < input.Start_Date)
{
    alert "End date cannot be before the start date.";
    cancel submit;
}
```

C.3 DS-03 Interview notification (planned)

```
// DS-03 | PLANNED - not implemented in the exported application
// Form: Interview_Schedule | Event: On Success
// Purpose: E-mail the candidate when an interview is scheduled
appRec = Application_Record[ID == input.Application];
stu = Student_Profile[ID == appRec.Student];
if(stu.Email != null)
{
    sendmail
    [
        from :zoho.adminuserid
        to :stu.Email
        subject :"Interview scheduled: " + input.Interview_Round
        message :"Date: " + input.Interview_Date + "<br/>Mode: " +
input.Interview_Mode
    ]
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application URL of the Internship and Placement Tracker in a web browser or in the Zoho Creator mobile app. Sign in with the Zoho account that the administrator invited. After signing in, the DashBoard and the sections permitted for your profile are displayed.

D.2 Instructions for Placement and HR Staff

- Open the Company Profile form and add each recruiting company with its contact and address details.
- Open the Internship Opportunity or Placement Opportunity form, select the company, and enter the details, eligibility, deadline and status.
- Open the Application Record form to record an application; select the student, the opportunity type and the opportunity, and update the application status as recruitment progresses.
- Open the Interview Schedule form, select the application, and enter the round, date, time and mode; after the interview, record the result and feedback.
- Use the grouped reports and the DashBoard (including its quick links) to review progress.

D.3 Instructions for Students

- Sign in and open the Student Profile form; your e-mail is filled in automatically and cannot be edited.
- Enter your register number, department, year, CGPA, graduation year and resume URL, and submit the form.
- Open the Student Skill form to add each skill with its proficiency level and certification details.
- Contact the placement office if expected information is not displayed; your access depends on your assigned profile.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	DHARINI BALAMURUGAN	Register No.	241501044
Project Title	DESIGN AND DEVELOPMENT OF AN INTERNSHIP AND PLACEMENT TRACKER USING ZOHO CREATOR	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	DHARINI BALAMURUGAN	Register No.	241501044
Department / Section	Artificial Intelligence and Machine Learning	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____